



Belajar Teknik ACL 🤞

ACL Ramadhan challenge 2026



Dates

28 Feb – 13 March

Length

14 days

Overview

serie latih tubi untuk para pelajar computer network yang ingin untuk mempelajari atau berlatih ACL.

slide dan latihan ini dibuat tanpa penggunaan AI untuk kegunaan orang ramai. Get creative dengan network design korang & have fun.

Topik

01

Block/allow ping between subnets with standard ACL.

28 Feb – 1 March



02

Permit HTTP from one host, deny others (extended ACL)

2 March – 3 March



03

Deny Telnet/SSH, allow specific IPs.

4 March – 5 March



04

IPv6 ACL to filter VLAN traffic.

6 March – 7 March



05

Troubleshoot broken ACL (wrong direction/order).

8 March – 10 March



06

Meet 5+ requirements (e.g., OSPF exempt, block ports).

11 March – 13 March



01

Block/allow ping between subnets with standard ACL.

28 Feb – 1 March

Network Overview

Device Role

R1 – Core Router – Site A

R2 – Distribution Router – Site B

R3 – Edge Router – Site C

SW1 – Layer 3 Switch – VLAN trunk

Requirements :

Two subnets that can/cannot ping each other based on standard ACL

ACL applied on router interface (inbound/outbound)

One subnet ALLOWED, one DENIED

IP Address Table

Device	Interface	IP Address	Subnet Mask	CIDR	Gateway
R1	Gi0/0 (to SW1)	10.10.1.1	255.255.255.128	/25	—
R1	Gi0/1 (to R2)	172.16.0.1	255.255.255.255	/32	—
R2	Gi0/0 (to SW2)	10.10.2.1	255.255.255.192	/26	—
R2	Gi0/1 (to R1)	172.16.0.2	255.255.255.255	/32	—
PC-A	NIC	10.10.1.10	255.255.255.128	/25	10.10.1.1
PC-B	NIC	10.10.1.20	255.255.255.128	/25	10.10.1.1
PC-C	NIC	10.10.2.10	255.255.255.192	/26	10.10.2.1
PC-D	NIC	10.10.2.20	255.255.255.192	/26	10.10.2.1



TOPIC 1
COMPLETED

Requirements

- One specific host permitted HTTP (port 80) to server
- All other hosts denied HTTP
- Extended ACL (source + destination + port)

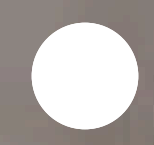
Extended ACL Rule (Applied on R1 Gi0/0 – Inbound)

```
1 access-list 110 permit tcp host 10.20.1.5 host 10.20.2.10 eq 80
2 access-list 110 deny    tcp any          host 10.20.2.10 eq 80
3 access-list 110 permit ip any any
```

IP Address Table

Device	Interface	IP Address	Subnet Mask	CIDR	Gateway
R1	Gi0/0 (LAN)	10.20.1.1	255.255.255.224	/27	—
R1	Gi0/1 (to R2)	172.16.1.1	255.255.255.255	/32	—
R2	Gi0/0 (Server LAN)	10.20.2.1	255.255.255.224	/27	—
R2	Gi0/1 (to R1)	172.16.1.2	255.255.255.255	/32	—
PC-A (Permitted)	NIC	10.20.1.5	255.255.255.224	/27	10.20.1.1
PC-B (Denied)	NIC	10.20.1.10	255.255.255.224	/27	10.20.1.1
PC-C (Denied)	NIC	10.20.1.15	255.255.255.224	/27	10.20.1.1
Web Server	NIC	10.20.2.10	255.255.255.224	/27	10.20.2.1





TOPIC 2
COMPLETED



03 Deny Telnet/SSH, allow specific IPs.

4 March - 5 March

Requirements

- Block Telnet (port 23) and SSH (port 22) from all except admin host
- Admin host IP specifically permitted
- Extended ACL protecting VTY lines

ACL Rule (Applied on VTY lines of R2)

```
access-list 120 permit tcp host 10.30.1.5 any eq 22
access-list 120 permit tcp host 10.30.1.5 any eq 23
access-list 120 deny tcp any any eq 22
access-list 120 deny tcp any any eq 23
access-list 120 permit ip any any
```

```
line vty 0 4
access-class 120 in
```

IP Address Table

Device	Interface	IP Address	Subnet Mask	CIDR	Gateway
R1	Gi0/0 (Admin LAN)	10.30.1.1	255.255.255.192	/26	—
R1	Gi0/1 (to R2)	172.16.2.1	255.255.255.255	/32	—
R2	Gi0/0 (User LAN)	10.30.2.1	255.255.255.128	/25	—
R2	Gi0/1 (to R1)	172.16.2.2	255.255.255.255	/32	—
Admin PC	NIC	10.30.1.5	255.255.255.192	/26	10.30.1.1
User PC-1	NIC	10.30.2.10	255.255.255.128	/25	10.30.2.1
User PC-2	NIC	10.30.2.20	255.255.255.128	/25	10.30.2.1
R2 Loopback0	Lo0	10.30.0.1	255.255.255.255	/32	—



TOPIC 3
COMPLETED

Requirements

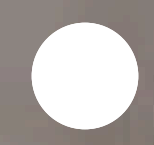
- IPv6 addresses assigned per VLAN
- IPv6 ACL blocks traffic between specific VLANs
- Applied on Layer 3 Switch SVI interfaces

IPv6 Address Table

Device	Interface / VLAN	IPv6 Address	Prefix	Gateway
SW1 (L3)	VLAN 10 SVI	2001:DB8:A:10::1	/64	—
SW1 (L3)	VLAN 20 SVI	2001:DB8:A:20::1	/64	—
SW1 (L3)	VLAN 30 SVI	2001:DB8:A:30::1	/64	—
R1	Gi0/0 (uplink)	2001:DB8:A:FF::1	/64	—
PC-VLAN10-A	NIC	2001:DB8:A:10::10	/64	2001:DB8:A:10::1
PC-VLAN10-B	NIC	2001:DB8:A:10::20	/64	2001:DB8:A:10::1
PC-VLAN20-A	NIC	2001:DB8:A:20::10	/64	2001:DB8:A:20::1
PC-VLAN30-A	NIC	2001:DB8:A:30::10	/64	2001:DB8:A:30::1

IPv6 ACL Rule (Block VLAN10 → VLAN20)

```
1  ipv6 access-list BLOCK_V10_TO_V20
2    deny  ipv6 2001:DB8:A:10::/64 2001:DB8:A:20::/64
3    permit ipv6 any any
4
5  interface vlan 10
6    ipv6 traffic-filter BLOCK_V10_TO_V20 out
```

TOPIC 4
COMPLETED

05

Troubleshoot broken ACL (wrong direction/order).

8 March – 10 March

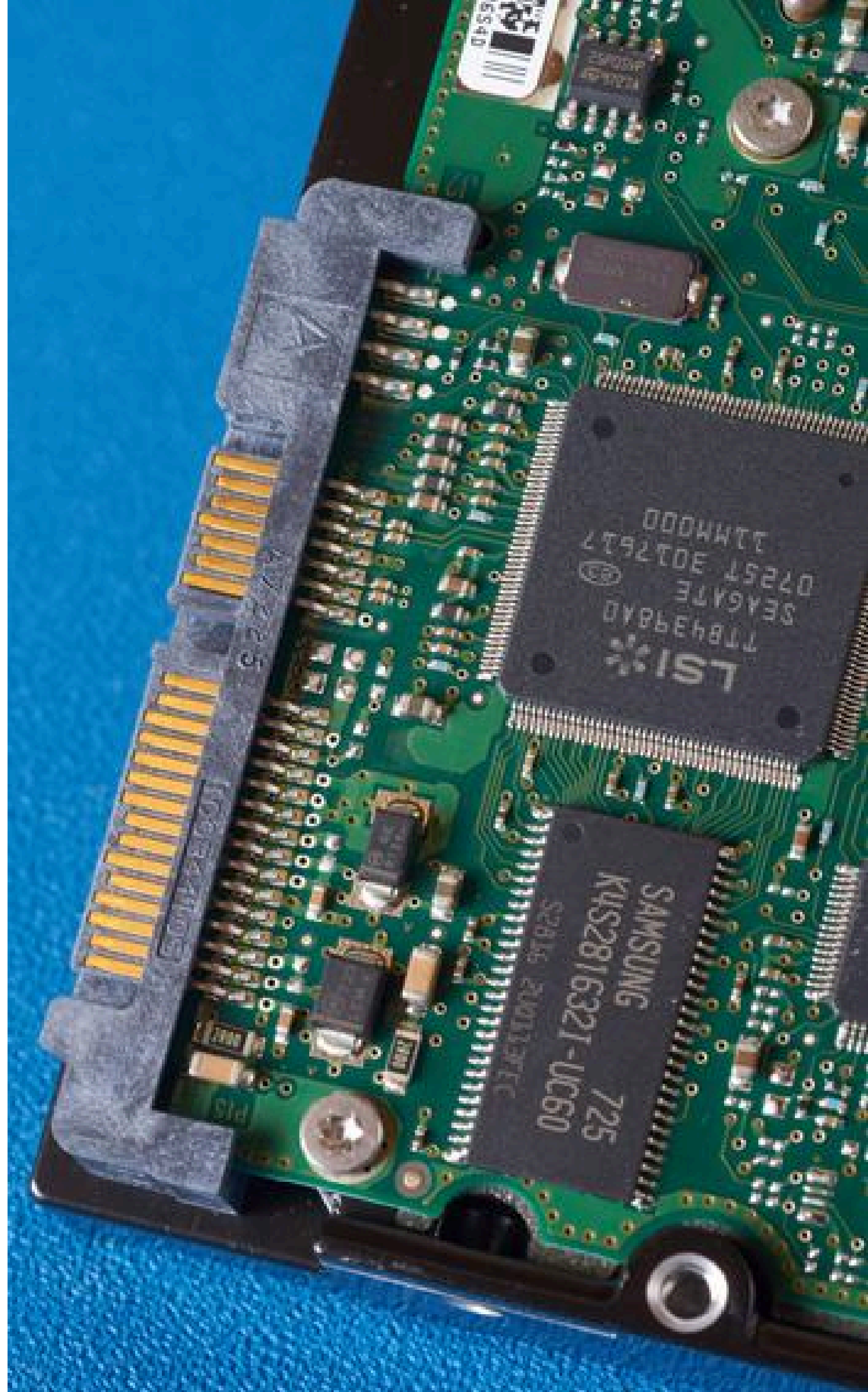
Requirements

- Intentionally misconfigured ACL (wrong interface direction or rule order)
- Student must identify and fix the issue
- Document: broken config → diagnosis → fixed config

IP Address Table

Device	Interface	IP Address	Subnet Mask	CIDR	Gateway
R1	Gi0/0 (LAN-A)	10.40.1.1	255.255.255.224	/27	—
R1	Gi0/1 (to R2)	172.16.3.1	255.255.255.255	/32	—
R2	Gi0/0 (LAN-B)	10.40.2.1	255.255.255.192	/26	—
R2	Gi0/1 (to R1)	172.16.3.2	255.255.255.255	/32	—
PC-A	NIC	10.40.1.5	255.255.255.224	/27	10.40.1.1
PC-B	NIC	10.40.1.10	255.255.255.224	/27	10.40.1.1
Server-1	NIC	10.40.2.5	255.255.255.192	/26	10.40.2.1

[GitHub Link](#)



06

Meet 5+ requirements (e.g., OSPF exempt, block ports).

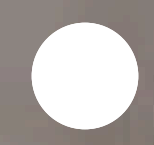
11 March – 13 March

Requirements (Must meet ALL 5+)

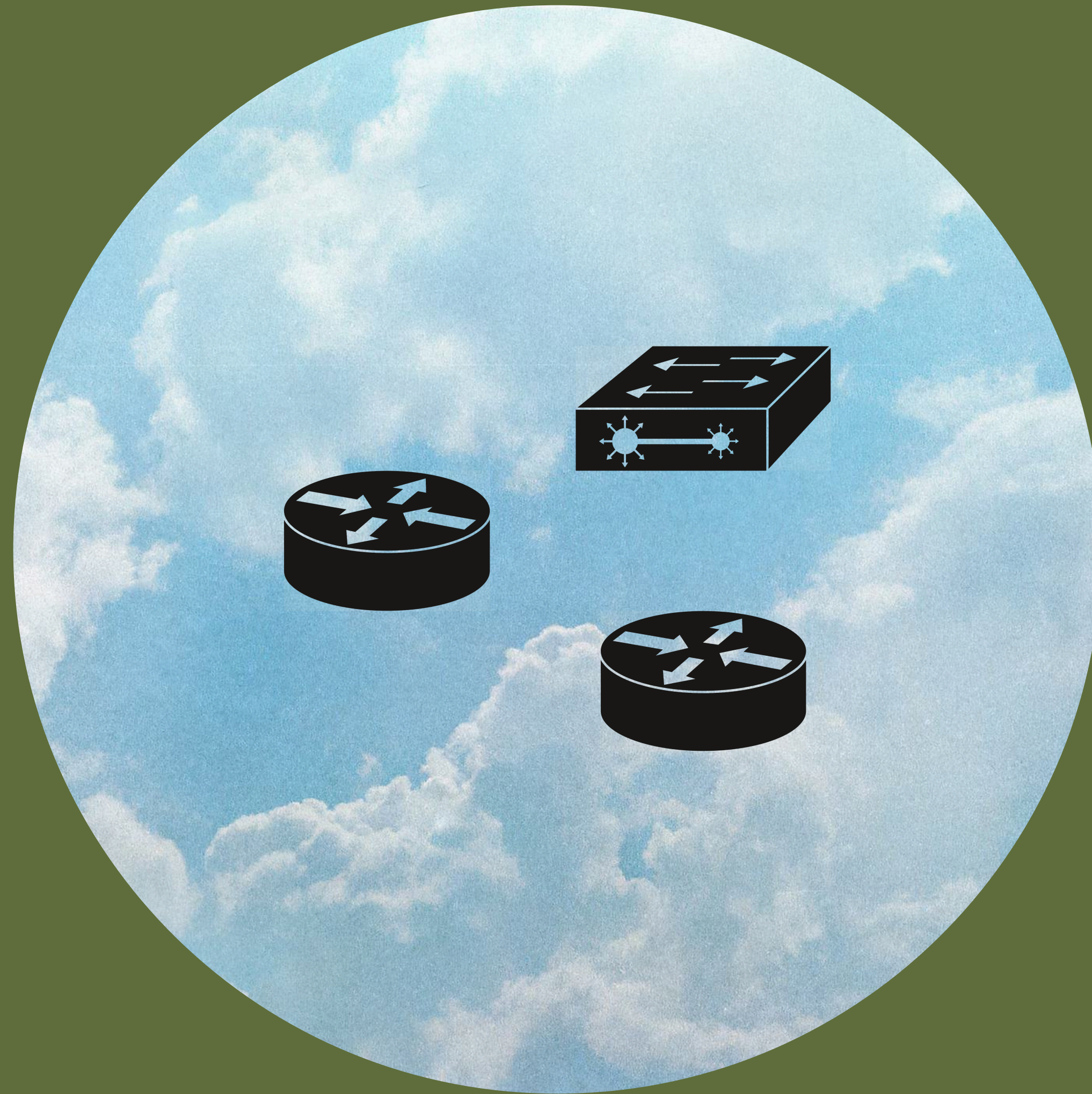
- OSPF routing traffic exempted from ACL filtering
- Block specific TCP ports (e.g., 8080, 445)
- Allow only one subnet to reach DMZ
- Deny ICMP from untrusted zone
- Permit management traffic only from admin subnet
- (Bonus) Rate-limit or log denied entry

IP Address Table

Device	Interface	IP Address	Subnet Mask	CIDR	Zone	Gateway
R1	Gi0/0 (Trust LAN)	10.50.1.1	255.255.255.128	/25	Trust	—
R1	Gi0/1 (to R2)	172.16.4.1	255.255.255.255	/32	Core	—
R1	Gi0/2 (DMZ)	10.50.3.1	255.255.255.224	/27	DMZ	—
R2	Gi0/0 (Untrust LAN)	10.50.2.1	255.255.255.192	/26	Untrust	—
R2	Gi0/1 (to R1)	172.16.4.2	255.255.255.255	/32	Core	—
R1 Lo0	Loopback	10.50.0.1	255.255.255.255	/32	Mgmt	—
R2 Lo0	Loopback	10.50.0.2	255.255.255.255	/32	Mgmt	—
Admin PC	NIC	10.50.1.5	255.255.255.128	/25	Trust	10.50.1.1
User PC-1	NIC	10.50.1.10	255.255.255.128	/25	Trust	10.50.1.1
Untrust PC	NIC	10.50.2.10	255.255.255.192	/26	Untrust	10.50.2.1
DMZ Server	NIC	10.50.3.10	255.255.255.224	/27	DMZ	10.50.3.1



TOPIC 6
COMPLETED



Have you ever estimate the budget of your network design ?

Router ???

Switch ???

Firewall ???

Server ???

Multilayer Switch ???

The end

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Narrated `Abdullah bin `Amr

Once the Prophet (ﷺ) remained behind us in a journey. He joined us while we were performing ablution for the prayer which was over-due. We were just passing wet hands over our feet (and not washing them properly) so the Prophet (ﷺ) addressed us in a loud voice and said twice or thrice: "Save your heels from the fire."